



LOCAL BEAM SEARCH



• A Local Search Algorithm in Artificial Intelligence •

WHAT IS LOCAL BEAM SEARCH?

Local Beam Search is a heuristic search technique that explores multiple candidate solutions at the same time.

Instead of following only one current state, it keeps the best k states and continuously generates their successors.

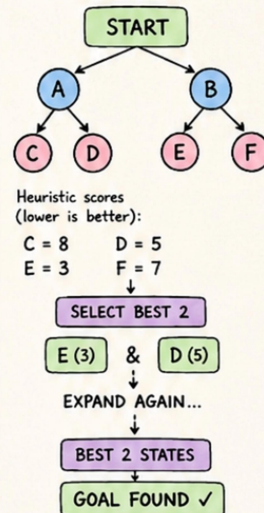
KEY IDEA:

"Keep the best k candidates
→ expand them → select the best k again."

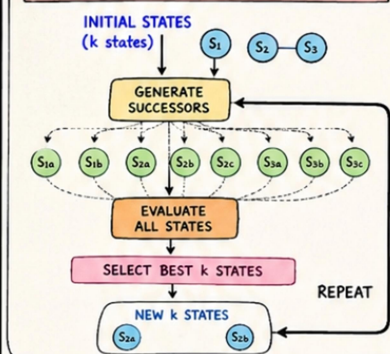
HOW DOES IT WORK?

- 1 START**
Choose k random initial states.
- 2 GENERATE**
Generate all possible successor states from the current k states.
- 3 EVALUATE**
Calculate the heuristic value of every successor.
- 4 SELECT**
Choose the best k states from all generated successors.
- 5 REPEAT**
Continue until a goal state is reached or no improvement is possible.

EXAMPLE: $k = 2$



VISUAL FLOW OF LOCAL BEAM SEARCH



PSEUDOCODE

Local-Beam-Search(k)

1. Generate k initial states
2. Repeat:
 - a. Generate all successors of current states
 - b. If a successor is a goal → return it
 - c. Select the best k successors
 - d. Replace current states with them
3. Return failure if no solution is found

LOCAL BEAM SEARCH VS HILL CLIMBING

	HILL CLIMBING	LOCAL BEAM SEARCH
Keeps	1 state	k states
Exploration	Explores one path	Explores multiple paths
Stuck in Local Optima	Can easily get stuck	More diverse exploration
Complexity	Simple & memory efficient	More computationally expensive

ADVANTAGES

- ✓ Explores multiple solutions simultaneously.
- ✓ Less likely to get stuck than simple hill climbing.
- ✓ Can find good solutions quickly.
- ✓ Useful for large search spaces.



LIMITATIONS

- ✗ May lose diversity if all k states become similar.
- ✗ Does not always guarantee the optimal solution.
- ✗ Performance depends on the value of k .
- ✗ Can still get trapped in local optima.



APPLICATIONS

- Route Optimization**
Finding efficient paths among many possible routes.
- AI Problem Solving**
Searching through large solution spaces.
- Scheduling**
Finding good combinations for timetables & resource allocation.
- Optimization Problems**
Useful when there are many possible candidate solutions.
- Game & Decision Systems**
Exploring several promising possibilities instead of only one.

KEY TAKEAWAY

Hill Climbing → "Follow the best path."

Local Beam Search → "Keep the best k paths."

The power of Local Beam Search comes from exploring several promising solutions at once.

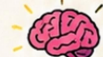


GROUP MEMBERS:

1. _____
3. _____
4. _____

SUBJECT: Artificial Intelligence

TOPIC: Local Beam Search



Program

```
function beamSearch(states : number[], k : number , goal : number ) : number{  
  while (true){  
    console.log("states : ", states)  
    if (states.includes(goal)){  
      console.log("Found goal : ", goal)  
      return goal;  
    }  
  
    const successors : number[] = [];  
  
    for (const state of states){  
      successors.push(state + 1);  
      successors.push(state - 1);  
    }  
  
    successors.sort((a, b) => Math.abs(a - goal) - Math.abs(b - goal));  
  
    states = successors.slice(0, k);  
  }  
}  
  
console.log(beamSearch([2, 5, 8], 3, 10));
```

Output :

[LOG]: "states : ", [2, 5, 8]

[LOG]: "states : ", [9, 7, 6]

[LOG]: "states : ", [10, 8, 8]

[LOG]: "Found goal : ", 10